



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Selected Topics in Cryptography

Lab 08: RSASSA-PSS

April 28, 2026

Develop the following exercises in pairs. You can use C/C++, C#, Java or Python to develop your source code, but please use only one of this languages to develop all the exercises. You **MUST USE any cryptographic library to implement RSASSA-PSS**.

1 Programming exercises

To develop the exercises of this section **you must consider the standard of NIST: SP800-186**

1. Implement a function to generate an RSASSA-PSS key pair whenever it is required. The private key and the public key must go to different textfiles. **You must store each key using base64**. The user must be able to choose the filenames.
2. Implement a function to do the signature generation with RSASSA-PSS. Please use a standard cryptographic hash function from the family SHA2 or SHA3. Your function must receive the name of the file containing the private key file d and the filename to be signed. Your function must return the signature
3. Design and implement a function to do the signature verification with RSASSA-PSS. Your function must read the textfile where the public key is stored, the filename containing the message m , and the signature. The function must return a boolean value: true if the signature is valid or false if it is not valid.
4. Your program must be able to do only one of the procedures: key pair generation, signature generation or signature verification.

2 Questions

Read the standard RFC-8017 to answer the following questions:

1. What does RSASSA-PSS mean?
2. What are the differences between RSASSA-PSS and RSASSA-PKCS1-v1_5? Which one is better and why?

3. Does the cryptographic library support RSASSA-PKCS1-v1_5? If not, look for a cryptography library that support it.
4. What are the functions or methods to implement RSASSA-PKCS1-v1_5? Specify the parameters required for each function.

3 Products

Every student must write her/his own report including the following information:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Answers to the questions of Section 2.